

# Abdominal cocoon: A rare case report of a small bowel obstruction.

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## Introduction

Sclerosing encapsulating peritonitis (SEP), or abdominal cocoon, is a remarkably rare entity. It consists of a fibrous layer or cocoon-like sac encasing, in most cases, the small bowel. The lack of specific symptoms makes preoperative diagnosis challenging.

## Case presentation

A 50-year-old patient, with no history of abdominal surgery or medical diseases, was admitted for symptoms of small bowel obstruction. On examination, the patient was underweight with a BMI of 18 kg/m<sup>2</sup>. Vital signs were stable. His abdomen was mildly distended and soft. Abdominal CT scan showed signs in favor of a left paraduodenal hernia with incarcerated small bowel loops. Intraoperatively, the small bowels were encapsulated in a cocoon-like structure formed by thick fibrous tissue associated with multiple adhesions. Careful adhesiolysis with complete resection of the membrane was performed. The postoperative course was uneventful. One year later, the patient was symptom-free.

## Discussion

SEP is thought to be a persistent inflammatory disorder due to a cause that remains unidentified. It is divided into two categories based on its etiology: primary and secondary. The most common mode of SEP revelation is bowel obstruction. Because there is no clear consensus on SEP management, the therapeutic approach typically follows the management strategy of bowel obstruction.

## Conclusion

SEP is an uncommon cause of bowel obstruction. Preoperative detection can be challenging in the absence of pathognomonic signs. Treatment is based on surgery, specifically entailing complete excision of the cocoon and adhesiolysis.

## Highlights

- Sclerosing encapsulating peritonitis (SEP), or abdominal cocoon, is a remarkably rare entity.
- Preoperative diagnosis of abdominal cocoon is challenging.
- A bottle gourd sign could guide the diagnosis of abdominal cocoon.
- It is crucial to improve the nutritional status of patients suffering from this condition.
- Surgical treatment is the keystone of management, based on complete excision of the cocoon and adhesiolysis.

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## Keywords

## Introduction

Sclerosing encapsulating peritonitis (SEP), or abdominal cocoon, is a rare condition, first reported in 1907 by Owtschinnikow [1]. It is defined as a thick fibro-collagenous membrane encasing the intestinal bowel and possibly other abdominal organs [2]. According to the underlying cause, SEP is classified into primary or idiopathic and secondary SEP. The clinical picture of primary SEP ranges from asymptomatic patients to those with recurring bouts of small bowel obstruction [3]. In this paper, we report a case of small bowel obstruction secondary to a primary SEP with initial imaging suggesting internal hernia. This work has been reported in line with the SCARE criteria [4].

## Case presentation

A 50-year-old patient, with no history of abdominal surgery or medical diseases, was admitted to our general surgery department for symptoms of small bowel obstruction evolving for 12 h. These symptoms included abdominal pain, nausea, vomiting, and abdominal distension. Throughout the previous year, the patient experienced similar episodes but never sought medical attention because of spontaneous resolution of symptoms. On examination, the patient had no fever, a blood pressure of 120 / 70 mmHg and a heart rate of 95 beats per minute. The patient was underweight with a BMI of 18 kg/m<sup>2</sup>. His abdomen was mildly distended and soft without tenderness or any other signs suggesting peritonitis. The rectum was empty upon examination. Laboratory findings showed an elevated white blood cell count without renal function impairment or electrolyte disorders. A biological nutritional assessment revealed low levels of serum albumin, while all other parameters fell within the normal range. An abdominal CT scan was performed, showing a bottle gourd sign with a dilated duodenum associated with a feces sign. The jejunum was collapsed. These findings were in favor of a left paraduodenal hernia with incarcerated small bowel loops (Fig. 1). There were no signs of ischemia or perforation. After admission, the patient was kept for 6 h under conservative management. He had appropriate fluid resuscitation and was kept nil per os under suction using a nasogastric tube. Parenteral nutritional supplements were administered. Evolution was favorable, with complete resolution of small bowel obstruction syndrome. The patient underwent a scheduled laparotomy ten days later. Intraoperative findings excluded the diagnosis of an internal hernia. The small bowels were encapsulated in a cocoon-like structure formed by a thick fibrous tissue associated with multiple adhesions (Fig. 2). Careful adhesiolysis with complete resection of the membrane were performed (Supplementary video). The review of the CT scan after surgery allowed for the identification of the Cocoon (Fig. 3). The postoperative course was uneventful. The anatomopathological report showed fibrous connective tissue with a collagen reaction. The diagnosis of primary SEP was retained after excluding tuberculosis and sarcoidosis based on the pathology report, the absence of a chronic drug use, and the absence of signs of cirrhosis according to CT and intraoperative findings. The patient remained asymptomatic after a 1-year follow-up.

## Discussion

SEP is a rare condition responsible for partial or complete encapsulation of intraperitoneal organs (including the small bowel, large bowel, and possibly abdominal viscera) [2]. Regardless of the underlying mechanism, SEP is thought to be a persistent inflammatory disorder due to a cause that remains unidentified. It starts with recurring latent peritonitis with no specific abdominal symptoms, evolving to sclerosis and membrane formation, thus forming a cocoon encasing the intestinal bowel [5]. Given that SEP is an uncommon disease, and its etiologies are diverse, the overall incidence and prevalence of this entity are yet to be determined [6]. According to the latest literature review conducted in 2022 by Chorti et al. [7], only 240 SEP cases were published throughout the international literature. SEP is divided into two categories based on its *primus movens*: primary (idiopathic) and secondary (acquired). Primary SEP was found to be less common, with no exact scientific evidence of an existing etiology underneath [2,8]. Studies conducted on peritoneal fibrosis and neoangiogenesis, suggested a vast myriad of possible mechanisms leading to primary SEP (e.g.: retrograde menstruation, retrograde peritonitis, omentum hypoplasia, and mesenteric vessel defect) [9,10]. More case studies are required to clarify the exact cause of idiopathic SEP. Secondary SEP's pathophysiology is currently better established as a local or systemic factor that sets in motion peritoneal inflammation. Among these factors, the most implicated in secondary

SEP is peritoneal dialysis, with its duration identified as an independent predictor of secondary SEP [11]. Other etiologies were also identified, such as systemic diseases (tuberculosis and sarcoidosis), long-term use of b-blockers, gynecological tumors (ovarian thecomas), cirrhosis, and transplanting organs [12]. In our case, these causes were ruled out and the diagnosis of primary SEP was retained. Fig. 1 Preoperative CT scan. (a) Axial view showing the bottle gourd sign with a dilated duodenum (yellow contour) associated with a feces sign (white arrows pointing the air bubbles in the duodenum). (b) Axial view showing the transition zone between the dilated duodenum (white star) and the collapsed jejunum (white circle). Fig. 2 Intraoperative view. A cocoon-like structure (white arrow) formed by thick fibrous tissue encapsulating the small bowels. Fig. 3 Preoperative CT scan reviewed after surgery. (a) and (b) axial views, (c) coronal view, and (d) sagittal view showing the cocoon capsule (white arrows). (c) The duodenum (white star) encircling the head of the pancreas (yellow square) is dilated, while the jejunum (white circle) is collapsed. Fig. 3

Preoperative CT scan. (a) Axial view showing the bottle gourd sign with a dilated duodenum (yellow contour) associated with a feces sign (white arrows pointing the air bubbles in the duodenum). (b) Axial view showing the transition zone between the dilated duodenum (white star) and the collapsed jejunum (white circle).

Intraoperative view. A cocoon-like structure (white arrow) formed by thick fibrous tissue encapsulating the small bowels.

Preoperative CT scan reviewed after surgery. (a) and (b) axial views, (c) coronal view, and (d) sagittal view showing the cocoon capsule (white arrows). (c) The duodenum (white star) encircling the head of the pancreas (yellow square) is dilated, while the jejunum (white circle) is collapsed.

Based on scientific papers, >50 % of SEP patients were admitted to emergency departments for symptoms of bowel obstruction [7]. This clinical presentation is the direct result of bowel compression and struggle in the limited space formed within the cocoon [13]. In most cases, these are recurrent symptoms that last several months before an urgent episode leads to emergency counseling. As for our case, our patient experienced multiple episodes of subocclusion over one year before seeking medical attention. In the largest survey conducted on primary SEP, the median symptom duration prior to emergency presentation was around 4 years, and because of this disease's chronic evolution, almost 75 % of patients were malnourished [9]. Abdominal masses and ascites were also described in some case series [14]. Even after a thorough investigation of the patient's history and a complete physical examination, due to the lack of specific symptoms in the early stages of SEP, preoperative diagnosis is challenging [15]. Until this day, no specific biomarker of SEP has been identified [16], with laboratory findings only in favor of infection or inflammation [17]. Given the limited number of documented cases, no consensus over the most effective approach for diagnosing SEP was ever reached, but contrast-enhanced computed tomography appears to be the imaging technique of choice [5]. The most distinctive radiological sign is the aggregation of small bowel segments at the midline, encased by a thick capsule (>2 mm) with a contrast-free border [18]. Other signs can be observed, such as the bottle gourd sign, ascites, increased fat density in the great omentum, and peritoneal calcifications. In our case, we found the bottle gourd sign preoperatively. The cocoon was identified after reviewing the images postoperatively. Despite an extensive clinical and radiological assessment, <30 % of SEPs are diagnosed in the preoperative phase, with mainly two conditions to consider as differential diagnoses: internal hernia and congenital peritoneal encapsulation [7].

The most common mode of SEP revelation is bowel obstruction, and because there is no clear consensus on SEP management, the therapeutic approach typically follows the management strategy of bowel obstruction. Conservative treatment based on fluid resuscitation, bowel rest, nasogastric decompression, and correct nutritional support is reserved for patients without signs of vital distress [19]. For patients suffering from SEP, nutritional status should not be taken lightly. It is crucial to improve the nutritional status of SEP patients since it may enhance their responsiveness to conservative therapy and avoid post-operative events such as infection and leaks [5]. Surgical management is the mainstay of treatment in SEP patients with clinical or radiological signs of ischemia or failure of conservative management. Surgical treatment consists of complete resection of the encapsulating fibrous membrane and adhesiolysis, as it

was done in our case. The recurrence risk is low if the membrane covering the bowel surface has been fully removed [20]. Treatment of the underlying cause is essential in cases of acquired SEP. Limited scientific proof has been brought to light concerning secondary SEP prevention, especially in peritoneal dialysis patients. Current human and animal experimental data suggests that tamoxifen and prednisone could be used for the prophylaxis of SEP [21].

## Conclusion

SEP is a rare disease with a pathogenesis based on inflammatory events. Its preoperative diagnosis requires a high index of clinical suspicion as there are no specific signs. A positive diagnosis is made in most cases based on operative findings. Medical conditioning, especially nutritional support, has a big impact on patient outcomes, but surgical treatment is the keystone of management, based on complete excision of the cocoon and adhesiolysis.

The following is the supplementary data related to this article. [Supplementary video](#) Intraoperative video showing the cocoon before the adhesiolysis and the result after dissection. [Supplementary video](#)

## Supplementary video

Intraoperative video showing the cocoon before the adhesiolysis and the result after dissection.

## Abbreviations

SEP Sclerosing encapsulating peritonitis

Sclerosing encapsulating peritonitis

## Ethical approval

Ethical approval was deemed unnecessary by our institutional ethical committee, as the paper is reporting a single case that emerged during normal practice.

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## Author contribution

Anis Hasnaoui: Conceptualization, Writing-Reviewing and Editing. Racem Trigui: writing-Original draft preparation. Sihem Heni: Data curation. Prakash V.A.K. Ramdass: Writing-Reviewing. All authors read and approved the final manuscript.

## Guarantor

Anis Hasnaoui

## Registration of research studies

None.

## Consent for publication

A written consent was obtained from the patient to publish this case report.

## Conflict of interest statement

The authors declare that they have no competing interests.

## References

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**Not applicable.**

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